PBEF Component 3: Participation Forecasting

Master Technical Briefing, Architecture Workflow, Dataset Audit & Performance Guide

Framework: Personalized Behavioral & Educational Framework (PBEF) | **Component Owner:** Naveeth Nawzer | **Date:** October 2026 | **Repository:** E:\PBEF

EXECUTIVE READINESS SCORECARD: Component 3 is **~90% complete** against the PP1 project rubric (exceeding the ~50% excellent target band). All 8 core script implementations, data contracts, observable annotation tools, zero-leakage window pipelines, 7 baseline/TCN models, and ONNX WebAssembly exports are completed, tested (11 unit tests passing), and verified with Code 0.

1. System Overview & Non-Diagnostic Ethical Contract

Core Research Objective: To determine how fine-grained interaction dynamics (touch latencies, inter-tap intervals, idle pauses, error runs, and prompt requests) on a learning tablet can forecast an imminent teacher-defined decrease in task participation **within the upcoming 20 seconds**, enabling proactive educator scaffolding while strictly preserving learner agency.

Item	Specification Contract & Design Boundary
Input Stream (Past 30s)	Recent 20–30s of fine-grained interaction events: response delays, screen touches, inter-touch pauses, error runs, and hint counts. Optional auxiliary head-pose and face-presence features with validity masks.
Target Horizon (Next 20s)	1. Binary future onset risk: Probability of an eligible teacher-defined participation decrease onset in [t, t+20s] (ground truth target). 2. Current participation state (5s window): Model classification of ACTIVE, PASSIVE_DECREASE, or REGULATING (model output, distinguished from upcoming ground truth).
System Outputs	Forecast risk score [0, 1], state classification, model confidence, validity mask, and expiration timestamp.
Owner Boundary	Component 3 supplies the risk prediction only. It strictly does NOT select educational activities or prompt interventions (delegated to C1, C2, and C4).
Strict Ethical Boundary	The model NEVER infers internal emotional, affective, or clinical states (e.g., 'boredom', 'meltdown', or 'autism severity'). Gaze shifts, processing pauses, and self-regulatory behaviors (stimming, rhythmic motor tapping) are classified as valid participation (REGULATING), never disengagement.

2. End-to-End System Workflow Architecture

The PBEF Component 3 processing pipeline operates across six strictly sequential stages designed to guarantee real-time client-side responsiveness and zero temporal leakage:

Stage	Component & Technical Role	Key Invariants & Guarantees
1. Event Logger	app/src/event_logger.ts Monotonic client-side telemetry embedded in tablet learning app.	Uses performance.now(). Logs touches, latencies, idle triggers, and stimulus lifecycles. Zero dropped frames.
2. Time Alignment	src/features/align_annotations.py Aggregates continuous asynchronous stream into 1-second bins.	Synchronizes interaction events with educator annotations and intervention markers on a unified millisecond timeline.
3. Window Builder	src/features/build_windows.py Extracts sliding temporal sequence windows (30s history / 20s horizon).	**Zero Calibration Leakage:** Window cutoff requires t >= 90s (initial 0-60s isolated exclusively for personal calibration plus 30s history).
4. Forecasting Models	src/models/train_forecaster.py & train_interaction_forecaster.py Deep 1D Temporal Convolutional Network (TCN).	Dual heads: Current-state Cross-Entropy + Future-onset BCE loss. Mode B achieves top ranking without cameras; Mode A gate adds regularization.
5. Event Evaluation	src/evaluation/evaluate_events.py Discrete onset event matching and false alert suppression.	Enforces a 15-second cooldown window following alerts. Computes usable advance lead time and false alerts per hour.
6. Browser Inference	src/models/export_onnx.py ONNX format export for client-side WebAssembly execution.	Dynamic batch axes. Numerical parity validated against native PyTorch ($< 10^{-4}$ maximum absolute error).

3. Current Status: What Has Been Done

All 8 scripts explicitly mandated by Page 3 of the *Component 3 Guide* have been implemented, tested, and documented. 11 client-side unit tests and master verification tests are passing:

Script / Module	Milestone	Implementation Details & Verification	Status
event_logger.ts	M2	Monotonic clock, stimulus lifecycles, touch buffers, idle detection. Passed 11 unit tests (app/src/event_logger.test.ts).	**DONE**
annotation_tool.tsx	M2	Interactive React suite with 5s ribbon, hotkeys (1-4, O, U, Space), inspector, and Cohen's Kappa agreement engine.	**DONE**
align_annotations.py	M3	Aggregates raw asynchronous interactions into fixed 1-second bins. Computes latencies and pause statistics.	**DONE**
build_windows.py	M3	Generates 30s input / 20s target windows. Strict participant/session isolation and t >= 90s calibration cutoff.	**DONE**
train_logistic.py	M4	Trains summary-statistic logistic regression and evaluates percentile-based deterministic threshold rules.	**DONE**
train_forecaster.py	M5	PyTorch 2-layer dilated TCN (32 ch) with multimodal reliability gate, class weighting, and joint dual-head loss.	**DONE**
evaluate_events.py	M6	Calculates event-level precision, recall, usable lead time, and false alerts per hour with 15s cooldown suppression.	**DONE**
export_onnx.py	M6	Exports PyTorch TCN to ONNX with dynamic batch axes. Validates numerical equivalence (diff $< 10^{-4}$).	**DONE**
train_interaction_forecaster.py	Mode B	Dedicated pure-interaction model (hit pacing & inactivity duration) achieving PR-AUC 0.4751 without cameras.	**DONE**
verify_all.py	Audit	Master verification suite executing unit tests, window checks, benchmark comparisons, and ONNX parity in sequence.	**DONE**

4. Dataset Audit & Decision Analysis: Any Other Dataset Needed?

Definitive Recommendation: NO additional public datasets are required or recommended for PP1. The project already integrates the gold-standard clinical autism dataset (Engagnition) and verified synthetic fixtures.

Dataset Candidate	Characteristics & Modalities	Assessment & Decision Rationale
Engagnition (Nature Sci Data 2024)	N=57 autistic children playing a serious game (37 analyzed, 6,705 windows total). Test split: 7 held-out participants (1,578 windows, 7.73% onset prevalence). LOPO across 6 onset participants.	**INTEGRATED & BENCHMARKED.** Serves as our primary clinical baseline. Complete LOPO cross-validation, participant concentration audit, and permutation ablation tests are complete.
DAiSEE (Video Affect Benchmark)	15 GB archive, 9,068 video clips, 112 neurotypical adult students watching e-learning lectures. Clip-level affect labels (Boredom, Confusion, Engagement, Frustration).	**EXCLUDED BY DESIGN.** 1. *Domain Mismatch:* Adult college students, not autistic children. 2. *No Interaction Data:* No touch events, reaction times, pauses, or errors. 3. *No Forecasting Target:* Static 10s clip labels cannot evaluate 20s future onset. 4. *Privacy:* C3 avoids mandatory video streaming.
Synthetic Verification Fixtures	9 sessions across 3 learners (L001-L003), 240s each, 1-second binning, 1,269 valid windows. Documented in data/synthetic/.	**FULLY VERIFIED.** Used strictly to mathematically verify zero calibration leakage, sliding window mechanics, and ONNX parity without clinical confounding.
Main D3 Local App Sessions	Real classroom tablet sessions (6-10 learners, 6-8 sessions each) with educators annotating 5-second windows and onset events.	**SCHEDULED POST-PP1.** The guide explicitly permits synthetic + public data for PP1 while local school pilot ethics approval and consent are obtained.

5. Work Performance & Quantitative Benchmark Results

Headline Finding: Mode B (interaction-only) matches or exceeds the full multimodal model on ranking metrics (PR-AUC 0.4751 vs 0.4233; ROC-AUC 0.9209 vs 0.9176), confirming that tablet touch dynamics carry the forecasting signal and that camera use is optional. The multimodal gate's contribution is calibration regularization, not additional predictive information.

Model Architecture	Input Features	PR-AUC	ROC-AUC	Brier Score	Scientific Finding
Mode B: Interaction Forecaster	Hit Pacing & Inactivity Only	**0.4751**	**0.9209**	0.2074	**Headline result:** Top ranking without cameras or wearables.
TCN Full Proposed	Multimodal Gate + Dropout	**0.4233**	**0.9176**	**0.0586**	**Regularized:** Gate acts as probability calibrator.
TCN Camera Fallback	Visual Channels Masked	0.4273	0.9193	0.0544	Resilient: Functional fallback (Δ = +6.2 pp on test window).
TCN Personal Baseline	Interaction+Motion Calib.	0.4709	0.9202	0.2524	High ranking, overconfident probabilities.
Population Logistic Reg.	Global Summaries	0.1725	0.5643	0.0714	+123% over chance baseline.
Prevalence Chance Baseline	Constant Prior	0.0773	0.5000	0.0755	Theoretical random baseline.
Personalized Logistic Reg.	Relative to Baseline	0.2027	0.3821	0.0703	Anti-predictive linear shift inversion.
TCN Population Baseline	Raw Sequences	0.0663	0.4565	0.1761	Overfits training participant signatures.

Leave-One-Participant-Out (LOPO) & Feasibility Findings:
Participant concentration is the central feasibility finding of PP1. On leave-one-participant-out evaluation across the 6 onset-bearing participants, the personalized TCN achieves mean PR-AUC **0.4915 ± 0.2789** and mean ROC-AUC **0.4256 ± 0.2061**, while population logistic regression achieves **0.5284 ± 0.2755** and **0.5842 ± 0.2911**. Personal-baseline normalization therefore does **not** replicate as a cross-participant advantage on this cohort. The pooled-split gain was attributable to participant-level concentration of onset events (P41 → P40/P35/P28), not to a transferable personalization mechanism. This is a feasibility boundary of the Engagnition cohort at N=6 onset-bearing participants, not a refutation of the mechanism — the in-situ pilot (D3) is the appropriate test.

- **Discrete Warning Detection: 9 of 11 true clinical episodes detected (81.82% recall)** [95% CI: 54.5%, 100.0%].
- **Advance Lead Time:** Mean = **8.67 seconds** (~8.7s benchmark average across test episodes, with 8s demo lead time from t=112s alert to t=120s onset).
- **Camera Fallback:** Masked prediction differs by 6.2 pp (61.2% → 67.4%), confirming functional fallback with the gate acting as a regularizer.

6. Running Guidance: Reproducibility & Commands

All procedures are executed from the repository root E:\PBEF using the isolated Python virtual environment:

Target Objective	Command Line Execution (PowerShell)	Expected Verification Output

1. Master Pipeline Verification	`.\.venv\Scripts\python.exe verify_all.py`	Executes unit tests, TS build, window extraction, 7-model benchmark, and ONNX parity. Exits Code 0.
2. Live PP1 Viva Demonstration	`.\.venv\Scripts\python.exe scripts/demo_heldout_window.py --pid P40`	Runs live ONNX inference on unseen autistic participant P40, shows advance warning risk and camera fallback (6.2 pp).
3. Mode B Interaction Forecaster	`.\.venv\Scripts\python.exe src/models/train_interaction_forecaster.py`	Trains and exports dedicated interaction-only ONNX model (Channels 0-1). PR-AUC: 0.4751, ROC-AUC: 0.9209.
4. Clinical Benchmark & Audit	`.\.venv\Scripts\python.exe scripts/run_audit_followup.py`	Executes LOPO cross-validation, permutation test, and bootstrap confidence intervals on Engagnition data.
5. Frontend Web Application	`cd app ; npm run dev`	Launches Vite React dashboard at http://localhost:5173/ with minimal white UI and deterministic session playback.

7. Remaining Work & Development Roadmap

The project is structured into immediate PP1 defense tasks and post-PP1 prototype integration tasks:

Phase	Action Item & Milestone	Deliverable & Next Step
Immediate (PP1 Defense)	**1. Viva Demo Rehearsal** Practice scripts/demo_heldout_window.py sequence.	Demonstrate 30s history -> 20s forecast -> camera-masked fallback (6.2 pp diff) -> teacher control.
Immediate (PP1 Defense)	**2. Failure Mode Defense** Defend participant concentration honestly.	Acknowledge LOPO cross-validation: population logistic regression outperforms personalized TCN across the 6 onset participants. Personalization does not replicate on this cohort; D3 pilot is the true test.
Immediate (PP1 Defense)	**3. Supervisor Scope Alignment** Confirm PP1 evidence boundary.	Present PP1_AUDIT_PACKAGE.md confirming synthetic verification + Engagnition clinical benchmark meets PP1 scope.
Post-PP1 (Prototype)	**4. Browser WebAssembly Runner** Integrate onnxruntime-web into React application.	Run model.onnx client-side in browser under WebAssembly execution provider.
Post-PP1 (Prototype)	**5. Device Latency Profiling** Measure client-side inference benchmarks.	Record p50 and p95 inference latency on tablet and desktop to substantiate offline readiness.
Post-PP1 (Prototype)	**6. Shared App Handover (C1, C2, C4)** Connect with peer component owners.	Provide docs/DATA_CONTRACT.md and event_logger.ts to activity and dashboard developers.
Post-PP1 (Prototype)	**7. School Pilot Ethics Clearance** Institutional ethics application for classroom trial.	Formal submission for 6-10 learner feasibility study following supervisor approval.